

Project - Protein Structure Feature Engineering Pipeline for Generative Deep Learning

This report presents a complete feature extraction, encoding, and decoding framework for protein structures designed for structure- based generative AI workflows. The pipeline converts raw protein structural information from a PDB file into a machine-learning-friendly representation suitable for generative AI workflows.

Objectives

1. Parse a PDB structure.
2. Extract meaningful structural and physicochemical features.
3. Design a deterministic feature encoder.
4. Design a decoder capable of reconstructing the original representation.
5. Validate reconstruction quality.
6. Demonstrate suitability for generative deep learning applications.

Protein Structure Parsing

BioPython's PDBParser was used to read the protein structure. The pipeline extracts alpha-carbon (CA) atoms for each residue. For every residue, the following information is collected:

- Residue type – X coordinate – Z coordinate

The CA representation provides a compact yet biologically meaningful abstraction of protein geometry.

Feature Engineering Strategy

A residue-level representation was selected because it balances relevance and computational efficiency. The extracted features include: - Amino acid identity – Three-dimensional spatial coordinates – Hydrophobicity score based on the Kyte-Doolittle scale. These features capture sequence information, structural information, and physicochemical properties simultaneously.

Graph Construction

Protein structures naturally form graphs. Each residue is represented as a node. Pairwise Euclidean distances are computed between residues. An edge is created when the distance between two residues is below 8 Å. Advantages: Captures local spatial interactions. Preserves neighborhood information. Suitable for graph networks and generative graph models.

Feature Encoding

The encoder transforms raw biological information into numerical vectors. Step 1: Residue Encoding One-Hot Encoding converts residue names into numerical vectors. Step 2: Coordinate Encoding StandardScaler normalizes X, Y, and Z coordinates. Step 3: Physicochemical Encoding Hydrophobicity values are appended as continuous features. Final Node Feature Vector: [One-Hot Residue Features | Normalized Coordinates | Hydrophobicity]. The encoded representation is deterministic and reproducible.

Feature Decoder

The decoder reconstructs the original representation. Residue Reconstruction : Inverse transformation of the One-Hot encoder restores residue identities. Coordinate Reconstruction: Inverse scaling reconstructs original Cartesian coordinates. Graph Reconstruction: Adjacency relationships are retained directly from the encoded graph representation. This enables interpretability and reversibility of the feature pipeline.

Validation Strategy

Mean Squared Error (MSE) is computed between original coordinates and reconstructed coordinates. A near-zero MSE demonstrates that the encoding-decoding process preserves structural information. Additional validation checks: -Residue identity consistency. – Coordinate reconstruction accuracy. – Graph connectivity preservation.

Generative AI Relevance

The designed representation is suitable for modern generative models because: - Node features can be used directly in Graph Neural Networks. – Spatial information is preserved. – Physicochemical characteristics remain interpretable. -Encoded tensors can be consumed by VAEs, Diffusion Models, Transformers, and Graph Generative Networks. The representation supports both structure generation and structure – conditioned molecule design workflows.

Conclusion

The project successfully implements a complete protein feature extraction, encoding, and decoding pipeline aligned with the requirements. The resulting representation preserves structural, spatial, and physicochemical information while remaining suitable for downstream generative deep learning workflows.